

Hans-Peter Marshall

Cryosphere Geophysics And Remote Sensing (CryoGARS) group
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Education

- 2005 Ph.D, Civil Engineering (Geotechnical), GPA: 3.9/4.0
Institute of Arctic and Alpine Research, University of Colorado at Boulder
Thesis: *Snowpack spatial variability: towards understanding its effect on remote sensing measurements and snow slope stability*
- 1999 B.S., Physics with honors, Geophysics Minor (1st in UW history), GPA: 3.6/4.0
University of Washington
Thesis: *Wet snow densification*
- 1994 Nathan Hale High School, Seattle, Washington, GPA: 3.9/4.0

Professional Experience

- 08/2018-present Project Scientist, NASA SnowEx Mission
- 08/2016-07/2018 Co-lead of Field Campaign, NASA SnowEx 2017 Mission
- 08/2013-present Associate Professor, Department of Geosciences and Center for Geophysical Investigation of the Shallow Subsurface (CGISS), Boise State University
- 07/2007-05/2020 Geophysical Engineer (Expert Consultant), U.S. Army Corps of Engineers Cold Regions Research and Engineering Lab (CRREL), Hanover, New Hampshire
- 08/2008-07/2013 Assistant Professor, Professor, Department of Geosciences and Center for Geophysical Investigation of the Shallow Subsurface (CGISS), Boise State University
- 04/2006-08/2008 Research Associate, Inst. of Arctic and Alpine Research (INSTAAR) University of Colorado at Boulder
- 12/2004-04/2006 Research Geophysical Engineer (GS-11), Student Career Experiences Program (SCEP) U.S. Army Corps of Engineers CRREL, Hanover, New Hampshire & Fairbanks, Alaska
- 01/2004-07/2004 Invited visiting scientist, Swiss Federal Institute for Snow and Avalanche Research (SLF), Davos, Switzerland
- 07/2000-12/2005 Research Assistant, INSTAAR, University of Colorado at Boulder
- 11/2002-12/2003 Avalanche Instructor, Alpine World Ascents, Boulder, Colorado
- 06/1995-07/2000 Undergraduate Research Assistant, Glaciology Group, University of Washington
- 08/1999-03/2000 Volunteer high school teacher, Old Crow, Yukon Terr., Canada
- Summers 1993-94 Roofer, Seattle, WA

Research Interests

Snow science and glaciology, using geophysics, engineering, and remote sensing
 Spatial variability of snow, snow and ice mechanics, snow slope stability, meltwater dynamics
 Computational statistics, geostatistics, compressed sensing, instrumentation development

Honors and Awards

2019	2018 WRR Editors' Choice Award: Hedrick et al., 2018
2018	NASA Robert H. Goddard Team Award: SnowEx Year 1 Organizing Team
2012	Timeless Award, University of Washington, 1 of 150 College of Arts and Sciences alumni awarded from UW history on 150th yr anniv. (1861-2011), 1 of 3 from Physics Department
2010-13	New Investigator Program Award, NASA, 1st in Idaho history
2010	Young Scientist Award, Cryosphere Section, American Geophysical Union
2009	Top Reviewer Award, Cold Regions Science and Technology (CRST)
2008	U.S. Young Scientist Award, International Union of Radio Science (URSI)
2002-05	N.A.S.A. Earth Systems Science Graduate Fellowship
2003	Backcountry Access Avalanche Education Support Grant
2003	Arctic Research Consortium of the U.S. (ARCUS) Award for Arctic Research Excellence, Hon. Mention
2002	University of Colorado Graduate School Student Travel Grant
2001-02	N.S.F. K-12 Engineering Outreach Fellowship
2002	Beverly Sears Graduate Research Grant (University of Colorado)
2001	American Avalanche Association Graduate Research Grant
2000	Li Tze Fong Memorial Fellowship, University of British Columbia (declined)
1999	University of Washington Bonderman Honors Travel Fellowship
1999	Volunteer of the Year, Arctic Canada, Frontiers Foundation
1997-99	University of Washington Mary Gates Undergraduate Research Grant
1999	Featured speaker, University of Washington Undergraduate Research Symposium
1995-98	N.A.S.A. Space Grant Undergraduate Research Program
1999,2001	N.S.F. Graduate Fellowship Program - Honorable Mention
1994-95,97-99	Dean's List, University of Washington
1994-95	U.W. Undergraduate Scholar Award (1994-95)
1994	Washington Athletic Club Student Athlete of the Year, Nathan Hale High School

Selected Refereed Publications [h-index=23, i10-index=41, citations=1588]

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| in-review | <p>Webb, R., Raleigh, M., McGrath, D., Molotch, N., Elder, K., Hiemstra, C., Brucker, L., and H.P. Marshall (in review). Within-stand boundary effects of snow water equivalent distribution in forested areas. <i>Water Resources Research</i></p> <p>Robertson, M., H.P. Marshall, Roberts, D., and Kunkel, M. (in review). Continuous snow water equivalent measurements with ultra-wideband impulse radar: A new sensor and time-delay method of processing downward-looking radar. <i>Hydrological Processes</i></p> <p>Deeb, E., Forster, R., and H.P. Marshall (in review). Estimating snow water equivalent (SWE) using L-Band interferometric synthetic aperture radar (InSAR). <i>Remote Sensing of Environment</i></p> |
| 2020 | <p>Hedrick, A. R., Marks, D., H.P. Marshall, McNamara, J., Havens, S., Trujillo, E., Sandusky, M., Robertson, M., Johnson, M., Bormann, K. J., and Painter, T. H. (2020). From drought to flood: A water balance analysis of the tuolumne river basin during extreme conditions (2015 – 2017). <i>Hydrol. Process.</i>, page doi:10.1002/hyp.13749</p> |
| 2019 | <p>McGrath, D., Webb, R., Shean, D., Bonnell, R., H.P. Marshall, Painter, T., Molotch, N., Elder, K., Hiemstra, C., and Brucker, L. (2019). Spatially extensive ground-penetrating radar snow depth observations during nasa’s 2017 snowex campaign: Comparison to in situ, airborne, and satellite observations. <i>Water Resources Research</i>, 55(11):10026–10036</p> <p>Lievens, H., Demuzere, M., H.P. Marshall, Reichle, R. H., Brucker, L., Brangers, I., de Rosnay, P., Dumont, M., Giroto, M., Immerzeel, W. W., Jonas, T., Kim, E. J., Koch, I., Marty, C., Saloranta, T., Schöber, J., and Lannoy, G. J. D. (2019). Snow depth variability in the northern hemisphere mountains observed from space. <i>Nature Communications</i>, 10(1):1–12</p> <p>Lund, J., Forster, R. R., Rupper, S. B., H.P. Marshall, Deeb, E. J., and Hashmi, M. (2019). Mapping snowmelt progression in the upper indus basin with synthetic aperture radar. <i>Frontiers in Earth Science</i>, 7:318</p> <p>Meehan, T. G., Marshall, H.-P., Deeb, E. J., and Shoop, S. A. (2019). Snowmicropenetrometer applications for winter vehicle mobility. ERDC 6.2 Boreal Aspects of Ensured Maneuver (BAEM) TR-19-14, U.S. Army Cold Regions Research and Engineering Laboratory</p> <p>Lewis, G., Osterberg, E., Hawley, R., Marshall, H. P., Meehan, T., Graeter, K., McCarthy, F., Overly, T., Thundercloud, Z., and Ferris, D. (2019). Recent precipitation decrease across the western greenland ice sheet percolation zone. <i>The Cryosphere</i>, 13:2797–2815</p> |
| 2018 | <p>Graeter, K. A., Osterberg, E. C., Ferris, D. G., Hawley, R. L., H.P. Marshall, Lewis, G., Meehan, T., McCarthy, F., and Birkel, S. D. (2018). Ice core records of West Greenland surface melt and climate forcing. <i>Geophysical Research Letters</i>, 45(7):3164–3172</p> |

- Hedrick, A., Marks, D., Kormos, P., **H.P. Marshall**, Havens, S., Bormann, K., and Painter, T. H. (2018). Direct insertion of NASA Airborne Snow Observatory-derived snow depth time-series into the iSnobal energy balance snow model. *Water Resources Research*, 54(10):8045–8063
- McGrath, D., Sass, L., O’Neel, S., McNeil, C., Candela, S. G., Baker, E. H., and **H.P. Marshall** (2018). Interannual snow accumulation variability on glaciers derived from repeat, spatially extensive ground-penetrating radar surveys. *The Cryosphere Discussions*, 12(doi.org/10.5194/tc-2018-126):3617–3633
- Kim, Y., Reck, T. J., Alonso-delPino, M., Painter, T. H., **H.P. Marshall**, Bair, E. H., Dozier, J., Chattopadhyay, G., Liou, K.-N., Chang, M.-C. F., and Tang, A. (2018). A Ku-band CMOS FMCW radar transceiver for snowpack remote sensing. *IEEE Transactions On Geoscience and Remote Sensing*, 66(5):2480–2494
- Hagenmuller, P., van Herwijnen, A., Pielmeier, C., and **H.P. Marshall** (2018). Evaluation of the snow penetrometer Avatech SP2. *Cold Regions Science and Technology*, 149:83–94
- McNamara, J. P., Benner, S. G., Poulos, M. J., Pierce, J. L., Chandler, D. G., Kormos, P. R., **H.P. Marshall**, Flores, A. N., Seyfried, M., Glenn, N. F., and Aishlin, P. (2018). Form and function relationships revealed by long-term research in a semiarid mountain catchment. *WIREs Water*, e1267(10.1002/wat2.1267)
- 2017 Mikesell, T., van Wijk, K., Otheim, L., **H.P. Marshall**, and Kurbatov, A. (2017). Laser ultrasound observations of mechanical property variations in ice cores. *Geosciences*, 7(3):47
- Lewis, G., Osterberg, E., Hawley, R., Whitmore, B., **H.P. Marshall**, and Box, J. (2017). Regional Greenland accumulation variability from Operation IceBridge airborne accumulation radar. *The Cryosphere*, 11(2):773
- Pegau, W. S., Garron, J., Zabilansky, L., Bassett, C., Bello, J., Bradford, J., Carns, R., Courville, Z., Eicken, H., Elder, B., Eriksen, P., Lavery, A., Light, B., Maksym, T., **H.P. Marshall**, Oggier, M., Perovich, D., Pacwardowski, P., Singh, H., Tang, D., Wiggins, C., and Wilkinson, J. (2017). Detection of oil in and under ice. *International Oil Spill Conference Proceedings*, 2017(1):1857–1876
- 2016 Evans, S., Flores, A., Heilig, A., Kohn, M., **Marshall, H.P.**, and McNamara, J. (2016). Isotopic evidence for lateral flow and diffusive transport, but not sublimation, in a sloped seasonal snowpack, Idaho, U.S.A. *Geophysical Research Letters*, 43(7):3298–3306
- Rutter, N., **Marshall, H.P.**, Tape, K., Essery, R., and King, J. (2016). Impact of spatial averaging on radar reflectivity at internal snowpack layer boundaries. *Journal of Glaciology*, 62(236):1065–1074
- Bradford, J. H., Babcock, E. L., **Marshall, H.P.**, and Dickins, D. F. (2016). Targeted reflection-waveform inversion of experimental ground-penetrating radar data for quantification of oil spills under sea ice. *Geophysics*, 81(1):WA59–WA70

- 2015 Heilig, A., Mitterer, C., Schmid, L., Wever, N., Schweizer, J., **Marshall, H.P.**, and Eisen, O. (2015). Seasonal and diurnal cycles of liquid water in snow-measurements and modeling. *Journal of Geophysical Research-Earth Surface*, 120(10):2139–2154
- Marshall, H.P.**, Hawley, R., and Tedesco, M. (2015). *Field measurements for remote sensing of the cryosphere*, chapter 14, pages 345–381. Remote Sensing of the Cryosphere. John Wiley & Sons, Ltd, ed. M. Tedesco
- Arenson, L., Colgan, W., and **Marshall, H.P.** (2015). *Physical, Thermal, and Mechanical Properties of Snow, Ice, and Permafrost*, chapter 2, pages 35–75. Snow and ice-related hazards, risks, and disasters. Elsevier, ed. Haeberli and Whiteman, Radarweg 29, PO Box 211, 1000 AE Amsterdam, Netherlands
- Hedrick, A., **Marshall, H.P.**, Winstral, A., Elder, K., Yueh, S., and Cline, D. (2015). Independent evaluation of the SNODAS snow depth product using regional-scale LiDAR-derived measurements. *Cryosphere*, 9(1):13–23
- 2013 Rott, H., Nagler, T., Prinz, R., Voglmeier, K., Fromm, R., Adams, M. S., Yueh, S., Elder, K., **Marshall, H.P.**, Coccia, A., Imbembo, E., Schuttemeyer, D., and Kern, M. (2013). AlpSAR 2012-13, a field experiment on snow observations and parameter retrievals with Ku-and X-band radar. Technical Report 722, European Space Agency Special Publication
- 2011 **Marshall, H.P.**, Schweizer, J., and Birkeland, K. (2011). Some recent advances in snow and avalanche science. *Cold Regions Science and Technology*, 69(2-3):119–121
- Koh, G., Lever, J., Arcone, S., **Marshall, H.P.**, and Ray, L. (2010). Autonomous fmcw radar survey of antarctic shear zone. *IEEE Publications, 13th International Conference on Ground Penetrating Radar*, DOI: 10.1109/ICGPR.2010.5550174
- 2008 **Marshall, H.P.** and Koh, G. (2008). FMCW radars for snow research. *Cold Regions Science and Technology*, 52(2):118–131
- Hardy, J., Davis, R., Koh, Y., Cline, D., Elder, K., Armstrong, R., **Marshall, H.P.**, Painter, T., Saint-Martin, G. C., Deroo, R., Sarabandi, K., Graf, T., Koike, T., and McDonald, K. (2008). NASA Cold Land Processes Experiment (CLPX 2002/03): Local Scale Observation Site. *Journal of Hydrometeorology*, 9(6):1434–1442
- 2007 **Marshall, H.P.**, Schneebeli, M., and Koh, G. (2007). Snow stratigraphy measurements with high-frequency FMCW radar: Comparison with snow micro-penetrometer. *Cold Regions Science and Technology*, 47(1-2):108–117
- 2005 **Marshall, H.P.**, Koh, G., and Forster, R. R. (2005). Estimating alpine snowpack properties using FMCW radar. *Annals of Glaciology*, 40:157–162
- 2004 **Marshall, H.P.**, Koh, G., and Forster, R. R. (2004). Ground-based frequency-modulated continuous wave radar measurements in wet and dry snowpacks, Colorado, USA: an analysis and summary of the 2002-03 NASA CLPX data. *Hydrological Processes*, 18(18):3609–3622

Archived Datasets

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| 2020 | H.P. Marshall and Gleason, A. (2020). SnowEx17 Senator Beck SnowMicroPen (SMP) Raw Penetration Force Profiles, Version 1. Technical Report doi: https://doi.org/10.5067/K7P8PB6EO1HO , NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA |
| 2018 | Elder, K., Brucker, L., Hiemstra, C., and H.P. Marshall (2018). Snowex17 community snow pit measurements, version 1. Digital media, https://doi.org/10.5067/Q0310G1XULZS , NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA |
| 2018 | Brucker, L., Hiemstra, C., H.P. Marshall , and Elder, K. (2018). Snowex17 community snow depth probe measurements, version 1. Digital media, https://doi.org/10.5067/WKC6VFMT7JTF , NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA |
| 2004 | Marshall, H.P. , Koh, G., and Forster, R. R. (2004). Ground-based frequency-modulated continuous wave radar measurements in wet and dry snowpacks, Colorado, USA: an analysis and summary of the 2002-03 NASA CLPX data. <i>Hydrological Processes</i> , 18(18):3609–3622 |

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